

Algebre Lineaire Cours 24

22/05/2025

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§1 Résumé (Réduction dans $\mathbb{R}^2$)

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, (x, y) \mapsto (\alpha x + \beta y; \gamma x + \delta y) \quad (1.1)$$

$$A_f = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \chi_f(X) = X^2 - \text{tr}(f)X + \det(A_f) \quad (1.2)$$

$$\Delta = \text{tr}(f)^2 - 4 \det(A_f) \quad (1.3)$$

§1.1 $\Delta > 0$:

$$\chi_f(x) = (x - \lambda_1)(x - \lambda_2) \quad (2)$$

Base propre $B = v_1, v_2$

$$v_1 \in \ker(f - \lambda_1 \text{id}), \mathfrak{I}(f - \lambda_2 \text{id}) \quad (3.1)$$

$$v_2 \in \ker(f - \lambda_2 \text{id}), \mathfrak{I}(f - \lambda_1 \text{id}) \quad (3.2)$$

$$[f]_B = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad (3.3)$$

§1.2 $\Delta = 0$

$$\chi_f(x) = (x - \lambda)^2 \quad (4)$$

Base de réduction $B = v_1, v_2; v_1 \in \ker(f - \lambda \text{id}), \mathfrak{I}(f - \lambda \text{id}), v_2 \notin \ker(f - \lambda \text{id}), \mathfrak{I}(f - \lambda \text{id})$

$$[f]_B = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \quad (5)$$

§1.3 $\Delta < 0$

$$\chi_f(x) = (x - \lambda)^2 + \mu^2, \mu \neq 0 \quad (6)$$

Base de réduction $B = v_1, v_2, v_1 \in \mathbb{R}^{2*}, v_2 = \frac{1}{\mu}(f - \lambda \text{id})(v_1)$

$$[f]_B = \begin{pmatrix} \lambda & -\mu \\ \mu & \lambda \end{pmatrix} = \sqrt{\lambda^2 + \mu^2} \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \quad (7)$$

§2 Puissances

Si B est une base de réduction pour $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ alors $R_f = P^{-1}A_fP, P = ([v_1]_{B_{\text{can}}} \ [v_2]_{B_{\text{can}}})$ et $A_f = PR_fP^{-1}, A_f^2 = PR_f^2P^{-1}$

$$A_f^n = PR_f^nP^{-1} \quad (8)$$

Proposition 2.1.

On a les 3 cas de réductions:

$$1. \quad \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}^n = \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix} \quad (9)$$

$$2. \quad \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}^n = \begin{pmatrix} \lambda^n & n\lambda^{n-1} \\ 0 & \lambda^n \end{pmatrix} \quad (10)$$

$$3. \quad \left[\rho \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \right]^n = \rho^n \begin{pmatrix} \cos(n\theta) & -\sin(n\theta) \\ \sin(n\theta) & \cos(n\theta) \end{pmatrix} \quad (11)$$

Preuve.

$$\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}^n = \left(\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right)^n \quad (12.1)$$

$$= \sum_{k=0}^n \binom{n}{k} \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}^k \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^{n-k} \quad (12.2)$$

$$= \sum_{k=n-1, n} \binom{n}{k} \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}^k \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^{n-k} \quad (12.3)$$

$$= n \begin{pmatrix} \lambda^{n-1} & 0 \\ 0 & \lambda^{n-1} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} \lambda^n & 0 \\ 0 & \lambda^n \end{pmatrix} \quad (12.4)$$

$$= \begin{pmatrix} \lambda^n & n\lambda^{n-1} \\ 0 & \lambda^n \end{pmatrix} \quad (12.5)$$

□

§2.1 Application Résoudre $m\ddot{x} = -kx$

$t \rightarrow x(t)$ On pose

$$X(t) = \begin{pmatrix} \dot{x}(t) \\ x(t) \end{pmatrix}, \dot{X}(t) = \begin{pmatrix} \ddot{x}(t) \\ \dot{x}(t) \end{pmatrix} = \begin{pmatrix} -\frac{k}{m}x(t) \\ \dot{x}(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{pmatrix} X(t) \quad (13)$$

On pose

$$X(t) = \exp\left(\begin{pmatrix} 0 & t \\ -\frac{kt}{m} & 0 \end{pmatrix}\right) X_0 \quad (14)$$

$$\frac{X(t)}{X_0} = \sum_{n=0}^{\infty} \frac{\begin{pmatrix} 0 & t \\ -\frac{kt}{m} & 0 \end{pmatrix}^n}{n!} = \sum_{n=0}^{\infty} P \frac{R^n t^n}{n!} P^{-1} \quad (15.1)$$

$$= P \left(\sum_{n=0}^{\infty} R^n \frac{t^n}{n!} \right) P^{-1} = P \exp(Rt) P^{-1} \quad (15.2)$$

$$\chi(x) = x^2 + \frac{k}{m} = x^2 + \mu^2, \mu = \sqrt{\frac{k}{m}} \quad (16)$$

$$R = \begin{pmatrix} 0 & -\mu \\ \mu & 0 \end{pmatrix} \quad (17)$$

$$Rt = \mu t \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad (18)$$

$$P = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{\frac{m}{k}} \end{pmatrix}, P^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{\frac{k}{m}} \end{pmatrix} \quad (19)$$

$$\exp(Rt) = \mathbb{1} + \mu t \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + \frac{\mu^2 t^2}{2} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} + \frac{\mu^3 t^3}{3!} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \frac{\mu^4 t^4}{4!} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \dots \quad (20.1)$$

$$= \begin{pmatrix} \cos(\mu t) & -\sin(\mu t) \\ \sin(\mu t) & \cos(\mu t) \end{pmatrix} \quad (20.2)$$

$$\frac{X(t)}{X_0} = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{\frac{m}{k}} \end{pmatrix} \begin{pmatrix} \cos(\mu t) & -\sin(\mu t) \\ \sin(\mu t) & \cos(\mu t) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{\frac{k}{m}} \end{pmatrix} \quad (21.1)$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{\frac{m}{k}} \end{pmatrix} \begin{pmatrix} \cos(\mu t) & -\sqrt{\frac{k}{m}} \sin(\mu t) \\ \sin(\mu t) & \sqrt{\frac{k}{m}} \cos(\mu t) \end{pmatrix} \quad (21.2)$$

$$= \begin{pmatrix} \cos(\mu t) & -\sqrt{\frac{k}{m}} \sin(\mu t) \\ \sqrt{\frac{m}{k}} \sin(\mu t) & \cos(\mu t) \end{pmatrix} \quad (21.3)$$

$$X_0 = \begin{pmatrix} v_0 \\ x_0 \end{pmatrix} \quad (22.1)$$

$$\begin{pmatrix} v \\ x \end{pmatrix} = \begin{pmatrix} \cos(\mu t)v_0 - x_0\sqrt{\frac{k}{m}}\sin(\mu t) \\ v_0\sqrt{\frac{m}{k}}\sin(\mu t) + x_0\cos(\mu t) \end{pmatrix} \quad (22.2)$$

$$\Rightarrow x(t) = x_0 \cos(\mu t) + v_0 \sqrt{\frac{m}{k}} \sin(\mu t) \quad (22.3)$$

$$\ddot{x}(t) = -v_0 \sin(\mu t) \sqrt{\frac{k}{m}} - x_0 \cos(\mu t) \frac{k}{m} = -\frac{k}{m} x(t) \quad (22.4)$$

$$m\ddot{x} = -kx - c\dot{x} \quad (23)$$

$$X(t) = \begin{pmatrix} \dot{x} \\ x \end{pmatrix}, \dot{X} = \begin{pmatrix} -\frac{k}{m}x - \frac{c}{m}\dot{x} \\ \dot{x} \end{pmatrix} = \begin{pmatrix} -\frac{c}{m} & -\frac{k}{m} \\ 1 & 0 \end{pmatrix} X(t) \quad (24)$$

$$X(t) = \exp(At)X_0 \quad (25.1)$$

$$\chi(x) = x^2 + \frac{c}{m}x + \frac{k}{m} \quad (25.2)$$

$$\Delta = \frac{c^2}{m^2} - 4\frac{k}{m} \quad (25.3)$$